Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2013 JC2 Preliminary Examinations

H2 BIOLOGY **9648/2**

Paper 2: Core (Mark Scheme)

Thursday 29 August 2013 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)
Answer **all seven** questions.
Write your answers in the spaces provided on the question paper.

Section B (Essay Question)
Answer **one** essay question.
Write your answers on the separate answer paper provided.
All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,
1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/13
3	/11
4	/12
5	/12
6	/11
7	/10
Total	/80

This document consists of **30** printed pages.
[Turn over

Section A

Answer **all** *questions*.

1. Fig. 1.1 shows the intracellular and extracellular events in the formation of a collagen fibril. Collagen fibrils are shown assembling in the extracellular space contained within a large infolding in the plasma membrane. Collagen fibrils can form ordered arrays in the extracellular space by further assembling into large collagen fibre.

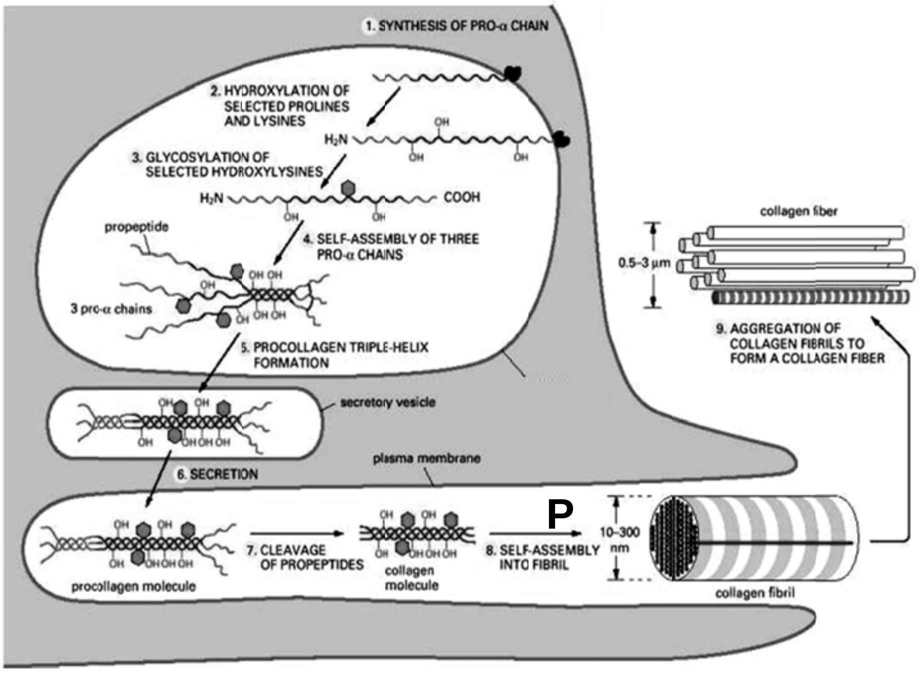


Fig 1.1

(a) **P** represents two organelles found in the eukaryotic cell. Identify these two organelles.

……………………………………………………………………………………………...[1]

1 rough endoplasmic reticulum
2 Golgi body

Examiners' comments:
More specific label must be given: ACCEPT bound ribosomes

Procollagen peptidase removes the terminal peptides of the procollagen to form collagen in Step 7 of Fig. 1.1.

(b) Explain the interaction between procollagen and procollagen peptidase.

……………………………………………………………………………………………...[2]

1 3D **conformation/structure** of procollagen molecule is complementary to
2 shape of active site of procollagen peptidase
3 form enzyme-substrate complex
4 ref. lock-and-key hypothesis / induced-fit hypothesis
5 contact amino acid in active site for brief bonding with substrate / via hydrogen bonds and/or ionic bonds
6 catalytic amino acids are brought into correct orientations in the active site / ref. inducing stress in bonds of substrate / ref. increasing substrate reactivity

(c) Suggest why assembly of collagen fibril is done outside the cell.

……………………………………………………………………………………………………[1]

1 collagen fibrils are too large (to be assembled inside the cell)
2 cannot be secreted out of cell
/ ref. affect internal cell environment

(d) Vitamin C is necessary in aiding the conversion of proline and lysine to hydroxyproline and hydroxylysine. A deficiency in Vitamin C could lead to scurvy, a disease displaying symptoms such as loss of teeth and easy bruising.

Collagen is an important matrix for the deposition of hard mineral crystals in the formation of bones. Certain forms of *osteogenesis imperfecta,* or brittle-bone disease, are caused by a single change of amino acid in the middle of the chain, from glycine to arginine.

With reference to the structure of collagen, account for the following:

(i) Bleeding gum and easy bruising in scurvy

……………………………………………………………………………………………………[4]

1 collagen is an important component in epidermis of gums and skin

2 lack of OH groups on proline
3 to form hydrogen bonds
4 to further stabilize tropocollagen structure

5 lack of hydroxylysine residues for formation of covalent bonds
6 between tropocollagen molecules forming collagen fibril
7 basic units of tropocollagen slide against each other

8 unable to form stable bundles of collagen fibres
9 leading to low tensile strength of collagen

Examiners' comments:
Candidates should answer the question more directly, instead of phrasing the answers in a indirect, reversed manner.
Order of packing: Tropocollagen (triple-helix) → collagen fibrils → collagen fibre
The terms "microfibrils" and "macrofibrils" are used to describe cellulose, NOT collagen!

(ii) weak and fragile bones in *osteogenesis imperfecta*

……………………………………………………………………………………………………[3]

1 (R group of) arginine is too large
2 to fit into the crowded centre of the triple helix (REJECT: α-helix)
3 unable to form hydrogen bonds between the –NH groups of arginine residues on each strand and –CO groups on the other two strands.
4 unable to form stable tropocollagen
5 results in low tensile strength
6 lack **stable** matrix for deposition of minerals in bone (REJECT: no matrix)

Tutor: reject general statements about change in 3D structure due to amino acid change

[Q1 Total: 11]

2.

(a) During semi-conservative DNA replication, both parental strands act as templates for synthesis of complementary daughter strands. One of the daughter strands, called the leading strand, is synthesized continuously. The other daughter strand, called the lagging strand, is synthesized discontinuously in the form of short fragments. The replicated DNA molecule consist of one new and one original strand of DNA.

(i) State 3 structural differences in the formation of the leading and lagging strands.

…………………………………………..…………………………………………..…[1½]

1 Presence of DNA ligase in lagging strand to ligate Okazaki fragments;
2 Presence of Okazaki fragments in lagging strand;
3 Presence of more than 1 primer/primase in lagging strand;
(**REJECT**: "no primer needed in leading strand". This is incorrect!)
4 Strands are synthesized in opposite directions;

(ii) Explain why the two new strands of DNA are synthesized in opposite directions.

…………………………………………...……………………………………..…..……[1]

1 DNA polymerase can only add nucleotides to the free <u>3' OH</u> end of elongating DNA strand
2 Template strands are antiparallel

(b) State three differences between the structures of DNA and RNA. [3]

DNA	RNA
1 Exists as a double-stranded helix	Single-stranded
2 Pentose sugar is deoxyribose	Pentose sugar is ribose
3 Bases are A, T, C and G	Bases are A, U, C and G
4 Ratio of A to T and G to C is always 1.	Ratio varies.

(c) In eukaryotic cells, the end-replication problem explains why normal somatic cells are limited in the number of mitotic divisions before they die out.

(i) Outline the end-replication problem in eukaryotic chromosomes.

………………………………………………………………………………..………….....…[3]

1 DNA polymerase can only add nucleotides to the free <u>3′ OH</u> end
2 of a pre-existing/elongating **polynucleotide** (not DNA, since primer is RNA)
3 primer (bound to the very end of the template strand) removed
4 but cannot be replaced with DNA
5 because no 3'-OH available (to which a DNA nucleotide can be added)
6 the <u>5' end</u> of the DNA becomes shorter relative to that of the previous generation

(ii) What are telomeres?

………[1]

1 ends of the <u>linear</u> DNA molecules
2 of <u>eukaryotic</u> chromosomes
3 ref. short **repeated** sequences

(iii) State the role of telomeres in relation to the end-replication problem.

………[½]

1 prevents genes from being lost as DNA shortens with each round of replication.

Examiners' comments:
Candidates must answer the questions specifically for the role of telomeres in relation to the end-replication problem. REJECT other roles of telomeres (i.e. protects the end of the chromosome from degradation by nucleases; prevents end-joining of chromosome ends which may lead to chromosomal mutations; prevents unintentional cell death).

Telomeres does not prevent DNA from shortening! It acts as a disposable buffer to protect the genes from erosion as DNA shortens with each round of replication.

Telomeres ≠ Telomerase!

(iv) Suggest why bacteria do not have telomeres.

………[1]

1 bacteria have circular DNA
2 no degradation of chromosome ends at each replication of DNA / no end-replication problem

In bacteria cells, binary fission differs significantly from the process of mitosis seen in eukaryotic cells.

(d) Explain the significance of binary fission in bacteria

………[2]

1 Ref. <u>asexual reproduction</u> for unicellular organism
2 Ensuring that offspring are genetically identical to the <u>parent</u>
3 Desirable alleles/traits are passed down

4 Rapid increase in cell numbers (under favourable conditions)
5 Important for survival of species

[Q2 Total: 13]

3.
(a) Fig. 3.1 shows two different viruses.

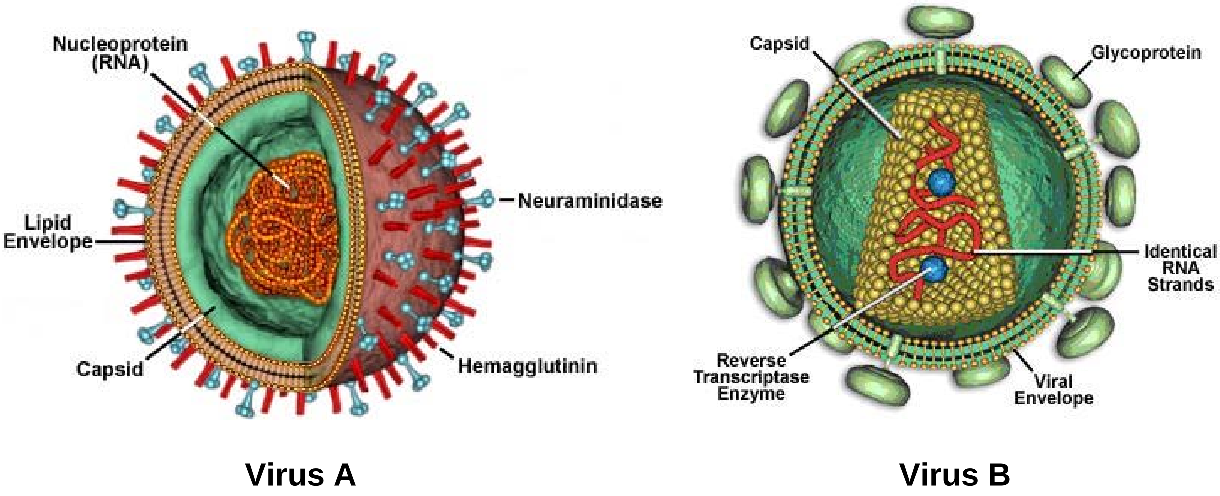


Fig. 3.1

(i) Relate the structure of virus A to its mechanism of release from the host.
………………………………………………………………………………………………[2]

1 Haemagglutinin binds to sialic acid (receptor) on host cells
2 Neuraminidase removes/cleaves sialic acid on viral envelope
3 to prevent agglutination/clumping of virus
4 to allow release of virus from host cell
/ to prevent virus from attaching to host cell during release

(ii) Contrast the reproductive cycles of the Virus A and Virus B. [3]

Virus A	Virus B
Enters host cell via **endocytosis**	Enters host cell via membrane **fusion** with plasma membrane of host cell
RNA genome is **not** reverse transcribed / negative-sense RNA is converted to positive-sense RNA	RNA genome is first **reversed transcribed** by reverse transcriptase into DNA
No integration of viral genome into DNA of host cell	Double-stranded DNA is **integrated** as <u>provirus</u> into DNA of host cell
No proteolysis**;**	Proteolysis of viral polyprotein to form mature viral proteins**;**
No latency phase	Latency phase

Examiners' comments:
REJECT <u>structural</u> differences e.g. reverse transcriptase vs no reverse transcriptase, enveloped virus vs not enveloped virus;
REJECT "metabolic machineries"
Budding (which happens for both influenza virus and HIV) ≠ Exocytosis !!

(b) One of the most common treatment methods for patients infected with virus B is the use of reverse transcriptase inhibitors. These inhibitors bind to an inhibitor-binding site (other than the active site) on the reverse transcriptase.

However, virus B often develop resistance against these inhibitors due to a single base substitution in the DNA sequence resulting in a change of an amino acid in a subunit of reverse transcriptase. The mutant protein is full-length and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

(ii) Explain why the mutated protein remains full-length and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

…………………………………………………………………………………………..………[2½]

<u>Remain full-length</u>

1 No frame-shift mutation
2 number of amino acids coded for remains the same

<u>Does not allow binding of inhibitors</u>

3 Tertiary structure/3D conformation of protein is altered
4 Inhibitor-binding site is no longer complementary to the 3D conformation of the inhibitor (no interaction between the inhibitor and reverse transcriptase)

<u>Retains reverse transcriptase activity</u>

5 But active site is not altered

(c) In recent years, vaccines have been developed for virus B. Vaccines can either be live but weakened forms of the virus, or its structural parts that are recognised by the immune system. This provides immunity to the disease.

However, vaccine development for virus B has become more challenging due to the virus's ability to undergo a high rate of antigenic drift.

(i) Suggest a reason for this high rate of antigenic drift and how it affects vaccine development.

…………………………………………………………………………………………..………[2½]

1 Reverse transcriptase lacks proof-reading ability
2 Unable to remove mismatched deoxyribonucleotides and insert the correct one (during DNA polymerisation)
3 Mutation in **genes** coding for protein/glycoprotein (gp120)
4 Produces altered (gp120) protein/glycoprotein
5 Antibodies induced by **earlier** strains of HIV / **previously synthesised** antibodies can no longer bind to protein/glycoprotein
6 Hence, new vaccines must be developed

Examiners' comments

- Vaccines ≠ Antibodies ≠ Antibiotics!
  - Vaccines = A vaccine typically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microbe, its toxins or one of its surface proteins. The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and "remember" it, so that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters.
  - Antibodies = Immunoglobulin proteins produced by the immune cells in response to foreign antigen (eg. virus or vaccines). Antibodies will bind to the antigen.
  - Antibiotics = chemicals that are used to kill bacteria. They have no effect on viruses.

(ii) A student tries to produce a vaccine by heating a human virus to 100°C for five minutes to inactivate its infectivity. Suggest why effective vaccines cannot be produced using the virus that has been heated to 100°C.

…………………………………………………………………………………...………………[1]

1 heating denatures the viral glycoprotein/antigens/capsid/envelope proteins;
2 unable to elicit specific antibodies from the immune system that recognizes the protein receptors/capsid from the viruses

Examiners' comments
Function of vaccines is to stimulate the production of new antibodies by the immune system. Some candidates mistakenly thought that vaccines will be targeted by existing antibodies.

[Q3 Total: 11]

4. Familial Dysautonomia (FD), is an autosomal disorder which affects the development of sensory neurons in the autonomic and sensory nervous system, resulting in various symptoms including the insensitivity to pain.

Mutation in another gene, transmembrane channel 1 gene (TMC1), results in a form of DFNB7 hearing loss which is also inherited in an autosomal manner.

A world-wide, long term genetic study on the possible associations between FD and hearing loss is implemented in 2003. Table 4.1 above shows the genotypes and phenotypes of some subjects in this study.

Subject number	Gene name	Genotype	Phenotype
1031	*FD*	Heterozygous	No FD
	TMC1	Heterozygous	Normal hearing
1050	*FD*	Homozygous recessive	FD
	TMC1	Homozygous dominant	Normal hearing
1088	*FD*	Homozygous dominant	No FD
	TMC1	Homozygous recessive	Hearing loss
1092	*FD*	Homozygous recessive	FD
	TMC1	Homozygous recessive	Hearing loss

Table 4.1

(a) Define the following terms:

(i) genotype

……………………………………………………………………………………..………………[1]

1 the genetic makeup of an organism;
2 the paired alleles that produces a phenotype;
3 can be either homozygous or heterozygous;

(ii) phenotype

……………………………………………………………………………………..………………[1]

1 physical and chemical characteristic of an organism;
2 depend on genotype of the organism and the environment;

(b) Subject 1031 and another subject listed in Table 4.1 are married and they have their first child in 2005. This cross between them theoretically resembles a "test cross".

(i) Deduce the subject number of subject 1031's wife from Table 4.1.

……………………………………………………………………………..…………..……[½]

Subject 1092;

Subject 1031 has a hearing impaired father with FD; his mother has normal hearing and does not suffer from FD (she is homozygous dominant for both gene locus).

A total of 200 couples with the same genotypes as subjects 1031 and his wife were monitored in this study. Information on their children (**total number of 186**) was collated 10 years after the implementation of this study and presented in Table 4.2.

Phenotype	**Number of children**
No FD, Normal hearing	59
FD, Normal hearing	35
No FD, Hearing loss	26
FD, Hearing loss	66

Table 4.2

(ii) Using the information give in Table 4.2, calculate the χ^2 and state the conclusions derived from this test.

……………………………………………………………………………………..………………[2]

1 23.42 (follow d.p. or sig.fig, whichever is applicable, of the χ^2-square table)
2 Since the calculated χ^2 value 23.42 > critical χ^2 value 7.82 at p = 0.05;
3 reject null hypothesis
4 The results of the χ^2 test suggest that there is a significant difference between the observed and the expected values at the 5% level
5 Any difference is not due to chance
6 The predicted phenotype ratio of 1:1:1:1 is incorrect;

$$\chi^2 = \Sigma\frac{(O-E)^2}{E} \qquad v = c - 1$$

where Σ = 'sum of…' O = observed 'value'
v = degrees of freedom E = expected 'value'
c = number of classes

A chi-squared table.

degrees of freedom	probability, p				
	0.10	*0.05*	*0.02*	*0.01*	*0.001*
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 = \Sigma\frac{(O-E)^2}{E}$$

$$= \frac{(59-46.5)^2}{46.5} + \frac{(35-46.5)^2}{46.5} + \frac{(26-46.5)^2}{46.5} + \frac{(66-46.5)^2}{46.5}$$

Degrees of freedom = c – 1
= 4 – 1 = <u>3</u>

(iii) Explain the discrepancy from the expected results.

………………………………………………………………………………….………………[3]

1 There is a larger number of parental phenotypes and smaller number of recombinant phenotypes;
2 There is no independent assortment of genes;

3 There is partial linkage of *FD* and *TMC1* genes;
4 the *FD* and *TMC1* genes are located on the same chromosome;

5 Recombinants are formed when;
6 crossing over occasionally breaks the linkage between genes on the same chromosome;
7 (Other scenario)If it is a normal dihybrid cross, the F2 generation results would show 4 types of phenotypes with a phenotypic ratio of 1:1:1:1
8 (Other scenario)If it is tight gene linkage, the F2 generation results would show 2 types of phenotypes with a phenotypic ratio of 1:1

(iv) Using the symbols provided, draw a genetic diagram of the cross between Subject 1031 and his wife. [2½]

Symbols:
F – dominant allele for no FD
f – recessive allele for FD
T – dominant allele for Normal hearing
t – recessive allele for hearing loss

Parental phenotypes:	No FD, Normal hearing	x	FD, Hearing loss
Parental genotypes:	$\frac{F\ T}{f\ t}$		$\frac{f\ t}{f\ t}$
Parental gametes:	FT ft Ft fT		ft

$\underbrace{\text{FT} \quad \text{ft}}_{\text{Parental gametes}} \quad \underbrace{\text{Ft} \quad \text{fT}}_{\text{recombinant gametes}}$

Punnett square
showing F_1 genotypes:

	FT	ft	Ft	fT
ft	$\frac{FT}{ft}$ □	$\frac{ft}{ft}$ △	$\frac{Ft}{ft}$ ♡	$\frac{fT}{ft}$ ○

Legend:

□ : No FD, normal hearing
△ : FD, hearing loss
♡ : No FD, hearing loss
○ : FD, normal hearing

F_1 phenotypes: $\underbrace{\text{No FD, normal hearing ; FD, hearing loss;}}_{\text{Parental}}$ $\underbrace{\text{No FD, hearing loss ; FD, Normal hearing}}_{\text{Recombinant}}$

1 Parental genotypes ;
2 Parental gametes;
3 F_1 genotypes;
4 F_1 genotypes correspond to phenotypes / legend for Punnett square;
5 Correct presentation of genetic diagram (circled gametes, Indication of recombinants and non-recombinant gametes, Indication of recombinants and non-recombinant phenotypes);

During gamete formation in Subject 1031's wife from a diploid germ cell, an error occurred once. As a result, her children will have 50% chance of having 3 copies of chromosome 9; and 50% of having only 1 copy of chromosome 9 in his/her cells.

(v) Explain how this could have happened.

…………………………………………………………………………………...………………[2]

1 Non-disjunction
2 homologous chromosomes fail to separate (in anaphase I);
3 so two copies of chromosome 9 will be found in 2 out of 4 oocytes / eggs/ gametes;
4 remaining 2 oocytes/eggs/gametes will have no chromosome 9;
5 upon fertilisation by fusion with the male gamete; (there is 50% chance of having 3 copies of chromosome 9 and 50% chance of having none in the offspring)

Examiners' comments:
Reject non-disjunction at anaphase II/separation of sister chromatids as an error that occurred once at meiosis II will result in 50% chance of normal (2 copies of chromosome 9); 25% chance of getting 3 copies of chromosome 9; 25% chance of getting only 1 copy of chromosome 9 after fusion with sperm

[Q4 Total: 12]

5.

(a) Fig. 5.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

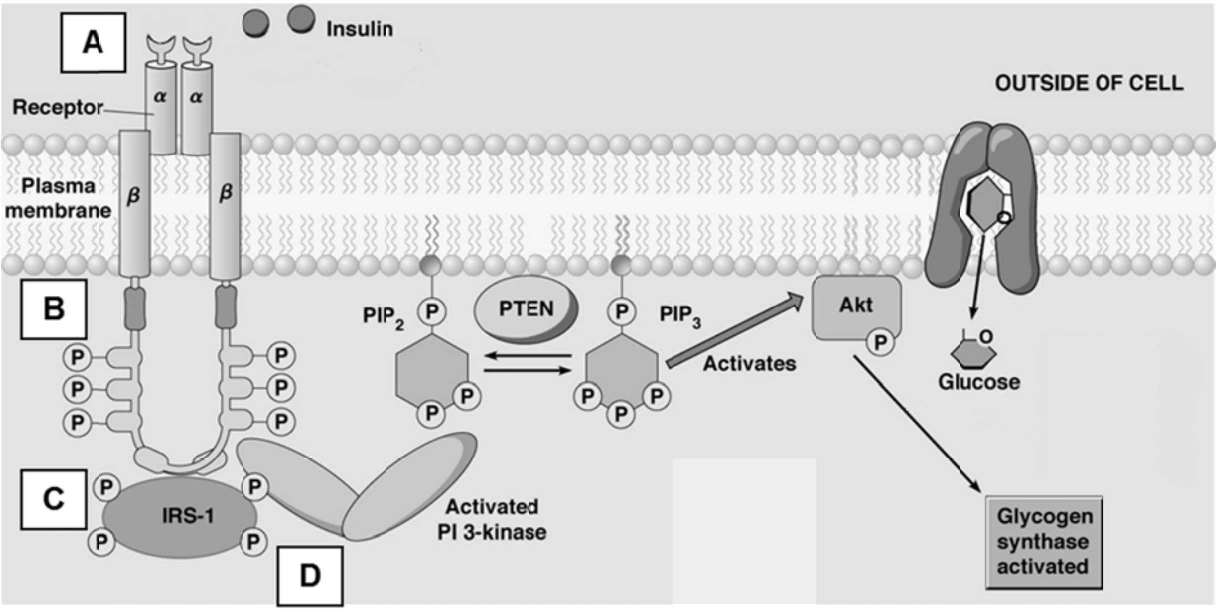


Fig. 5.1

(i) State the source of the insulin molecule in Fig 5.1.

……………………………………………………………………………………………....[1]

1 β-cells of the islets of Langerhans
2 of the pancreas

(ii) With reference to the structure of the plasma membrane, describe how the structural features of receptor tyrosine kinase enables it to be embedded as a transmembrane protein.

……………………………………………………………………………………………....[3]

1 **hydrophobic** region of protein consist of amino acid with non-polar R groups)
2 interacts with **hydrophobic core** of membrane
3 consisting of **fatty acid tail** of phospholipid bilayer
4 hydrophilic region of protein consist of amino acids with polar/hydrophilic R groups
5 interacts with **hydrophilic** region of membrane
6 consisting of **phosphate heads** of phospholipid bilayer

(iii) With reference to Fig. 5.1, describe the events occurring in stages **A** to **D**.

………………………………………………………………………....……………………....[3]

1 insulin receptor exists as a dimer prior to the binding of insulin
2 insulin bind to insulin receptor
3 ref. complementary in shape to the ligand binding site of the insulin receptor
4 results in the activation of the tyrosine kinase domains in the cytoplasmic tail.
5 each receptor <u>phosphorylates</u> the tyrosine residues (at the cytoplasmic tails) of the other receptor, hence activating it
6 auto-crossphosphorylation
7 receptor <u>phosphorylates</u> and activates **IRS-1**
8 which in turn <u>phosphorylates</u> and activates **PI 3-kinase**

Examiners' comment

The question asks for a description of stages A, B C. Candidates must use the information given in the diagram i.e. specifically referring to IRS-1, PI 3-kinase and phosphorylation rather than just proteins, tyrosine kinases, relay proteins, activation.

(iv) Based on your knowledge, describe the events that lead to an increase in the number of glucose transporters in the cell surface membrane.

………………………………………………………………………....……………………....[1]

1 **movement** of **vesicles** containing glucose transporters (GLUT4 transporters) towards the cell surface membrane.
2 **fusion** of vesicle membrane with cell surface membrane results in **incorporation** of glucose transporters on the cell membrane.

(v) A tissue may, over time, lose its responsiveness to insulin, even though insulin concentration remains unchanged. Suggest possible reasons for this decrease in responsiveness.

………………………………………………………………………....……………………....[1]

1 Mutation in the gene coding for insulin receptor / change in the shape of insulin receptor as cell/tissue ages
2 Mutation in the gene coding for insulin receptor / change in the shape of Glut4 transporters as cell/tissue ages

3 Fewer insulin receptors are synthesised / ref. lower transcription rate of gene that codes for insulin receptor
4 Fewer Glut4 transporters are synthesised/incorporated / ref. lower transcription rate of gene that codes for Glut4 transporters

(b) An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 5.2 shows the various experimental systems that are prepared in an attempt to synthesize ATP.

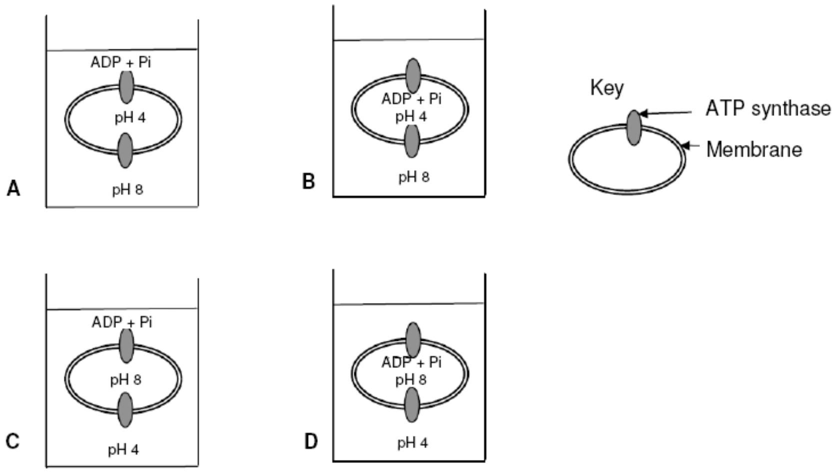


Fig. 5.2

Which experimental system (A to D) shown in Fig. 5.2 would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer.

…………………………………………………………………………………..…………..……[3]

1 Experimental system A
2 pH within the membrane is lower (pH 4 as compared to pH 8 outside the membrane)
3 Concentration of H^+ ions is higher within the membrane (than outside the membrane)
4 ref. proton gradient
5 tendency for the H^+ ions to diffuse out of the vesicle
6 via ATP synthase;
7 ATP is formed using ADP and P_i found <u>outside</u> the membrane

Examiners' comments
pH 4 represents a higher concentration of H^+. Some candidates made careless mistakes when relating pH and H^+ concentrations.
Some candidates merely describe the chloroplast without reference to the diagram. This will earn them minimum marks.
Only ATP synthase can be seen in the diagram; no ETC or stalked particles are drawn.

[Q5 Total: 12]

6.

(a) During an action potential, the permeability of the cell-surface membrane of an axon changes. Fig 6.1 shows changes in permeability of the membrane to sodium ions (Na^+) and to potassium ions (K^+) during a single action potential.

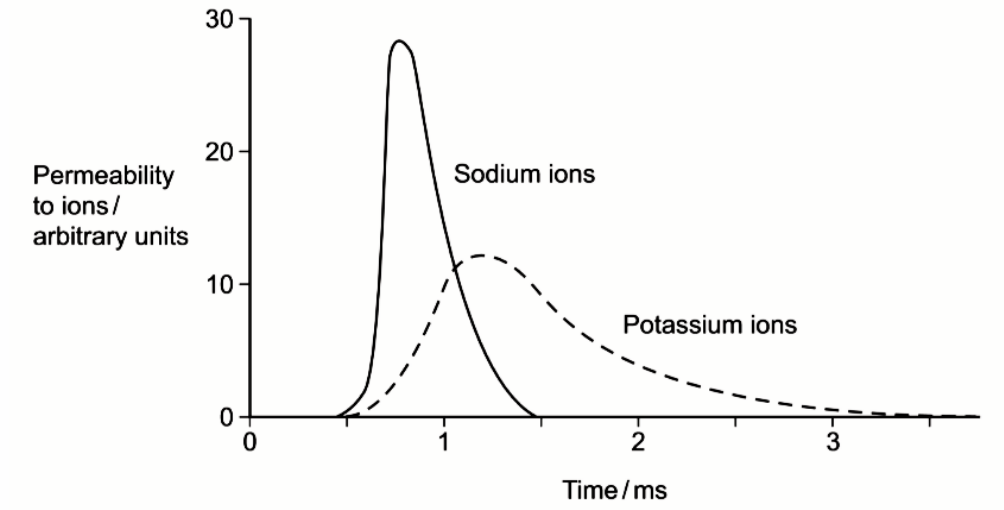


Fig. 6.1

(i) Explain the shape of the curve for sodium ions between 0.5 ms and 0.7 ms.

…………………………………………………………………………………………..……….[2]

1 Between 0.5 ms and 0.7 ms, permeability to Na^+ ions increases from 0 a.u. to 28 a.u. (acceptable range 23 – 29 a.u.)
2 Some voltage-gated Na^+ ion channels open
3 Na^+ ions diffuse into the neurone
4 Changes membrane potential/makes inside of axon less negative/positive/depolarisation/ reaches threshold
5 All voltage-gated Na^+ ion channels open
6 More Na^+ ions diffuse into the neurone

Examiners' comments

The type of ion channels (eg. ungated, voltage-gated, ligand-gated) must be clearly stated.
In this question about action potential occurring in the axon, the ligand-gated ion channels are NOT applicable.

(ii) During an action potential, the membrane potential rises to +40mV and then falls. Using information from the graph, explain the fall in membrane potential.

…………………………………………………………………………………………..……….[2]

1 Repolarisation occurs**;**

2 Voltage-gated Na^+ ion channels (in the membrane) are closed**;**
3 decreasing/stopping diffusion of Na^+ ions into neurone**;**
4 ref. 0.7 ms to 1.5 ms, permeability to Na^+ ions decreases from 28 a.u. to 0 a.u. (acceptable range: 27-29 a.u.)

5 voltage-gated K^+ ion channels (in the membrane) open**;**
6 increasing diffusion of K^+ ions out of neurone**;**
7 ref. 0.5 ms to 1.2 ms, permeability to K^+ ions increases from 0 a.u. to 12 a.u. (acceptable range: 11-13 a.u.)

(iii) After exercise, some ATP is used to re-establish the resting potential in axons. Explain how the resting potential is re-established.

…………………………………………………………………………………………..……….[1]

1 <u>Na^+- K^+ pump</u> on membrane
2 Pumps 3Na^+ out and 2K^+ in neurone (by active transport)
3 restores the <u>unequal distribution/concentration gradient</u> of Na^+ ions and K^+ ions (in the intracellular and extracellular fluids at resting potential)

(b) The black mamba is a poisonous snake which produces a toxin.

(i) The length of the section of DNA that codes for the complete toxin is longer than the mRNA used for translation. Explain.

……………………………………………………………………………...…………...……[½]

1 DNA contains introns/non- coding bases
/ mRNA only contains exons/coding bases

(ii) The black mamba's toxin kills prey by inhibiting the enzyme acetylcholinesterase at neuromuscular junctions that controls muscles involved in breathing. Explain how this inhibition affects breathing.

……………………………………………………………………………………………...……[2]

1 Acetylcholine not broken down
/ stays bound to receptor
2 Na^+ ions **continue** to enter
/ **continued** depolarisation
/ Na^+ channels **continuously** open
3 action potentials generated/impulses fired **continuously**;
4 Muscles involved in breathing **remain** contracted/cannot relax;

Examiners' comments:
'Muscles contract' alone is not sufficient; muscles <u>remain</u> contracted.
Inability to breath is not due to lack of action potential generation. Students should read the question carefully to note that acetylcholinesterase is inhibited.

(c) Dopamine is a neurotransmitter released by dopaminergic neurons for the transmission of nerve signals important for motor control. Dopamine binds to and activates its receptors located on the post-synaptic membrane.

Fig 6.2 shows a crystallography of the dopamine receptor.

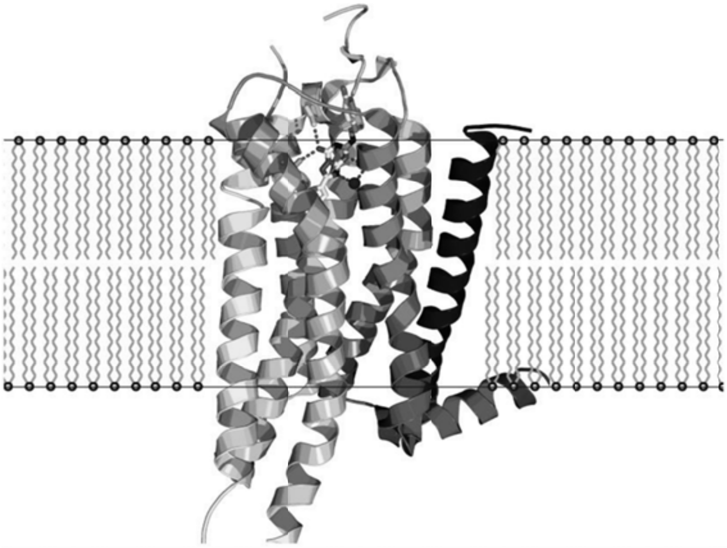


Fig 6.2

(i) With reference of Fig. 6.2, identify the type of receptor which the dopamine receptor belongs to.

……………………………………………………………………………………………………[½]
G-protein coupled receptor

Excessive neurotransmission of dopamine is associated with schizophrenia, a clinical mental disorder. To relieve this disorder, antipsychotic drugs are developed to bind to and inactivate the post-synaptic dopamine receptors.

Fig 6.3 shows the cell signalling pathway when dopamine binds to its receptors.

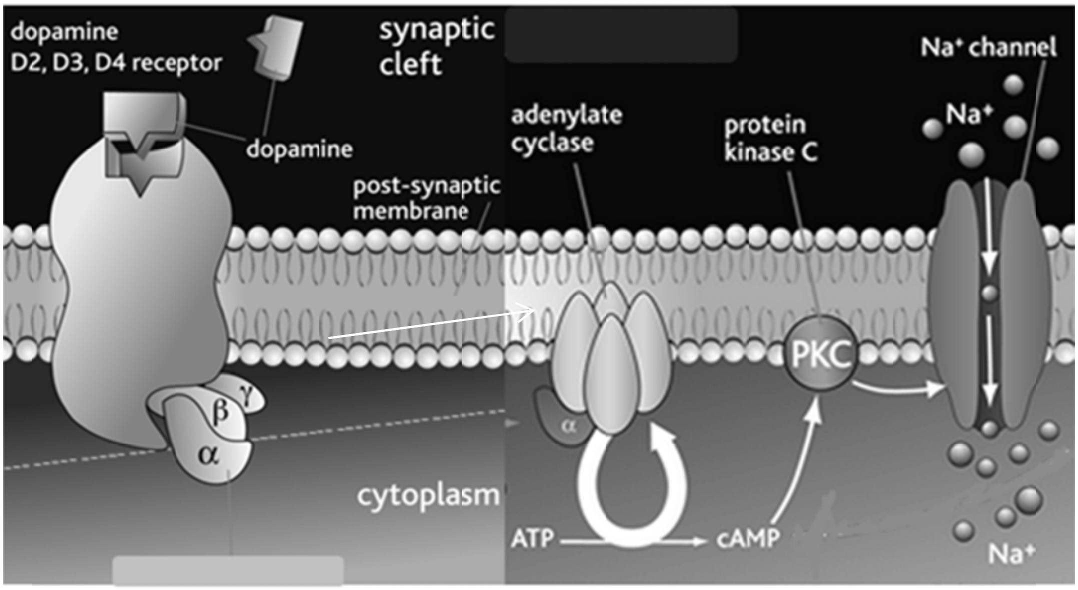


Fig 6.3

(ii) With reference to Fig 6.3, briefly describe how binding of the drugs to the receptor affects downstream signalling pathway.

………………………………………………………………………………………………..[3]

1 Prevents conformation change in receptor (hence no activation of receptor)
2 G protein unable to bind to receptor
3 GDP not displaced by GTP
4 G protein remains inactive / α-subunit of G protein does not dissociate
5 and bind to / activate adenylyl cyclase
6 No conversion of ATP to cAMP.
7 No activation of protein kinase C
8 No opening of Na^+ ion channels

[Q6 Total: 11]

7. Fishes in the genus *Anisotremus* comprise of ten described species which occur predominantly on coral reefs and subtropical rocky reefs in the Neotropics.

(a) Bernardi *et.al.* did a molecular phylogenetic study on such fishes in that area. Results are shown in Fig. 7.1.

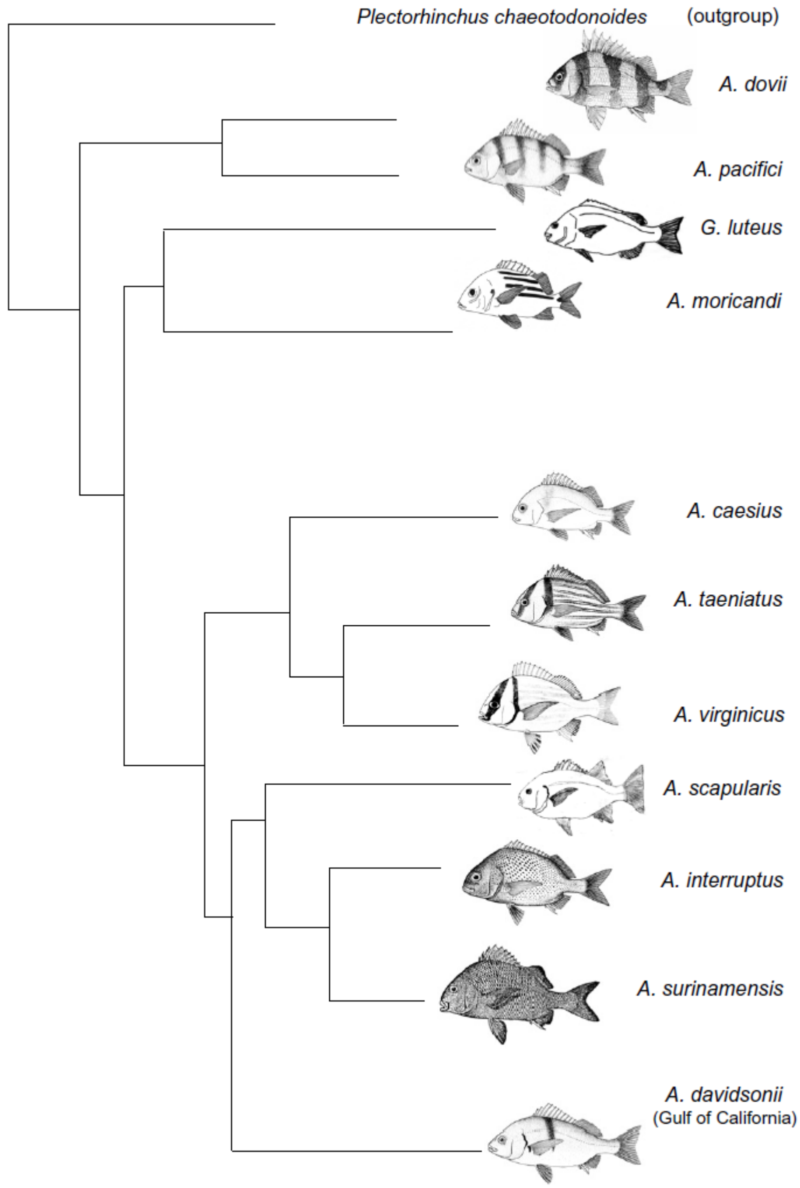


Fig. 7.1

(i) Explain how molecular methods can be used to elucidate the evolutionary relationships of the different species of *Anisotremus* fishes.

……………………………………………………………………………………………………[2]

1 Neutral mutations do not confer any selective advantage or disadvantage;
2 Neutral mutations are accumulated at a relatively constant rate for any particular gene and use as a molecular clock;
3 compare DNA sequence of a particular **common** gene (between different species of fish);
/ comparison/alignment of **homologous** genetic sequences

4 E.g. rRNA genes, cytochrome c genes, hypervariable regions of mtDNA
5 number of mutation in genetic sequence is used to calculate the length of time since divergence
/ % sequence homology indicates degree of evolutionary closeness

(ii) Explain why it may be more reliable to construct a phylogenetic tree of the ten species of *Anisotremus* using molecular data instead of morphological comparisons.

…………………………………………………………………………………………….……….[1]

1 Quantifiable **;**
protein, nucleic acid sequence data are precise and accurate and easy to quantify / convertible to numerical form for mathematical and statistical analysis**;**

OR

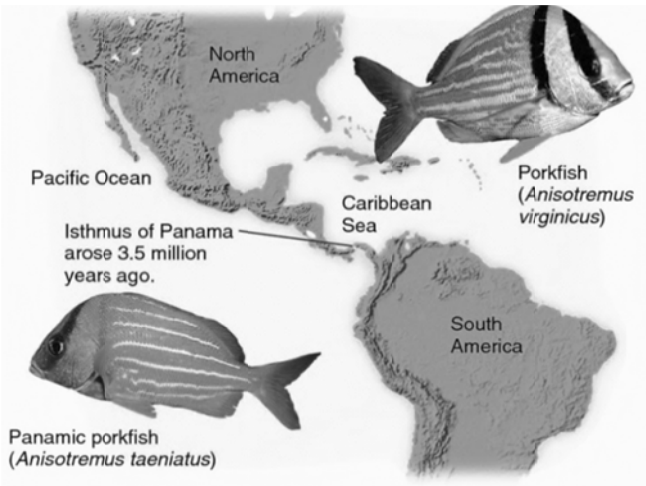
2 Objective**;**
based strictly on heritable material
/ can be easily described in an unambiguous manner
/ some morphological characteristics may be analogous / ref. convergent evolution**;**

(iii) With reference to Fig. 7.1 and molecular homology, comment on the evolutionary relationship between *A. taeniatus* and *A. virginicus*.

………………..……………………………………………………………..………...…..…[1]

1 *A. taeniatus* and *A. virginicus* are **closely related**;
2 share a (recent) **common ancestor**;
3 **high percentage homology** in DNA sequence alignment;

(b) *A. taeniatus* is found in the Pacific Ocean, whereas *A. virginicus* is found in the Caribbean Sea. These two species were derived due to the formation of the Isthmus of Panama about 3.5 million years ago. Before that event, the waters of the Pacific Ocean and Caribbean Sea mixed freely.



(i) Explain how the formation of the **I**sthmus of Panama results in the emergence of *A. taeniatus* and *A. virginicus*.

………………………………………………………………………………………………….[4]

1 <u>geographical isolation</u>
/Isthmus of Panama is a physical barrier
/ ref. <u>allopatric speciation;</u>
2 <u>disruption to gene flow</u> in the ancestral population
/ <u>no interbreeding</u> between the organisms in the Pacific Ocean and Caribbean Sea;
3 processes of <u>natural selection</u> and <u>genetic drift</u> occur;

4 <u>genetic variations</u> exist within each sub-population;
5 <u>different selection pressures</u> in Pacific Ocean and Caribbean Sea;
6 list 1 eg. food availability/ salinity /temperature/ different predators;
7 individuals with a <u>selective advantage</u> in the particular environment survived till reproductive age and pass on their alleles to offspring;
8 <u>change in allele frequency</u> of gene pool;
9 accumulation of genetic differences over time;
10 different populations ultimately cannot interbreed to produce viable, fertile offspring;

(c) Explain why it is impossible for evolution to occur at the individual level.

………………………………………………………………………………………………….[2]

1 Evolution = changes in allele frequencies in a gene pool of a population over time;

2 There must be variation (in a population) before selection can take place;
3 Individuals are selected for or against by natural selection;
4 Individuals can only pass <u>alleles</u> to the next generation;
5 It is the populations that actually evolve

[Q7 Total: 10]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

8 **(a)** Distinguish between the concepts of repressible and inducible systems of gene regulation. [4]

(b) Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [10]

(c) Describe how gene amplification can occur and the significance of gene amplification in development [6]

[Total: 20]

OR

9 **(a)** Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]

(b) Explain, with two named examples, how the environment may affect the phenotype. [6]

(c) Explain how biogeography and the fossil record support the evolutionary deductions based on homologies. [7]

[Total: 20]

8(a) Distinguish between the concepts of repressible and inducible systems of gene regulation. [4]

	Repressible system	Inducible system
1	**Repressible enzymes** are produced	**Inducible enzymes** are produced
2	Involved in **anabolic** (biosynthesis) pathways	Involved in **catabolic** (hydrolysis) pathways
3	Repressors are synthesized in **inactive form**	Repressors are synthesized in **active form**
4	Transcription of structural genes is ***normally* switched on**	Transcription of structural genes is ***normally* switched off**
5	Pathway end-product (tryptophan) serves as **co-repressor** that signals repression of the *trp* operon	Substrate molecule serves as **inducer** (allolactose) that signals the induction of the *lac* operon
6	Operon is repressed to **prevent over production** of pathway product	Operon is induced to **stimulate production** of enzymes necessary for breakdown substrate

8(b) Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [10]

<u>For every comparison:</u>
½ mark for correct *comparison*
½ mark for correct *information*

Chromosomal Level (max 2)

	Prokaryotes	Eukaryotes
1.	Prokaryotic DNA not organised into chromatin / not associated with histones	Eukaryotic DNA is complexed with histones and other proteins to form chromatin / associated with histones
2.	DNA demethylation/methylation and histone acetylation/deacetylation cannot occur	DNA and histone modification can occur, resulting in **conversion between** euchromatin and heterochromatin
3.	DNA sequences, eg. **promoters** and **operators**, serve as the on/off switch	Structure of chromatin – **euchromatin**, ready to be transcribed, or **heterochromatin** and not available – is the major <u>on/off switch</u> for gene regulation / Ease of transcription – DNA made accessible to RNA polymerase and other regulatory proteins

Transcriptional Level (max 2)

Prokaryotes	Eukaryotes
4. One RNA polymerase consisting of five subunits / All RNAs synthesized by the same RNA polymerase;	Three different RNA polymerases, each containing 10 or more subunits; / Three different classes of RNA each synthesized by a different RNA polymerase (i.e. mRNA, tRNA, rRNA)
5. Simple regulatory sequence: Transcriptional regulatory protein / Regulator protein binds to DNA-binding sites upstream of the cluster of structural genes to regulate initiation of transcription.	Complex regulatory sequence: More extensive interaction between upstream DNA sequences and protein factors involved to stimulate and initiate transcription. In addition to **promoters, enhancers and silencers** control rate of transcription.
6. **Related genes are transcribed together as operons** / only 1 promoter / polycistronic mRNA	No operon / each gene has own promoter / monocistronic mRNA

Post-transcriptional Level (max 2)

Prokaryotes	Eukaryotes
7. Translation is often coupled to transcription / Transcription and translation take place in the same cellular compartment simultaneously.	No direct coupling of transcription and translation / mRNA must pass across nuclear envelope before translation in the cytoplasm. RNA transcript is not free to associate with ribosomes prior to the completion of transcription.
8. Primary transcripts are the actual mRNAs / no post-transcriptional modification	Primary transcripts undergo processing to produce mature mRNAs – **methylated guanosine cap at the 5’ end, poly-A tail at the 3’ end, splicing**
9. Lower stability of transcript / degradation within seconds or minutes / mRNAs shorter half life to rapidly respond to environmental changes	Higher stability of transcript / prevent transcript degradation / mRNAs longer half-life remaining much longer to orchestrate protein synthesis prior to their degradation by nucleases in the cell

Translational level (max 2)

Prokaryotes	Eukaryotes
10. Control at this level is unlikely; due to simultaneous transcription and translation	Control at translational level: phosphorylation of ribosomal translation initiation factors / negative translational control through regulatory proteins / cytoplasmic elongation of poly (A) tails / mRNA degradation / RNA interference and microRNA
11. mRNAs have multiple ribosome binding sites / direct the synthesis of several different polypeptides	mRNAs have only one start site / direct synthesis of only one kind of polypeptide

Post-translational Level (max 2)

Prokaryotes	Eukaryotes
12. no/minimal post-translational modifications occur	Post-translational modifications determine the functional abilities of the protein
13.	Proteolysis: <u>Processing</u> eukaryotic polypeptides to yield <u>functional</u> protein molecules e.g. cleavage of pro-insulin to form the active insulin hormone
14.	Chemical modification of proteins to yield functional protein molecules
15.	Phosphorylation of proteins to increase or decrease its function
16.	Transportation of proteins to target destinations in the cell where it functions is mediated by signal sequences at N-terminus of some proteins. Once transported to destination, signal sequence is enzymatically removed from the proteins
17.	Ubiquitination marks protein for degradation, ref to ubiquitin & proteasome

8(c) Describe how gene amplification can occur and the significance of gene amplification in development . [6]

1 gene amplification is the the (selective) replication of certain genes
2 increases the number of templates for transcriptions
/ results in increased gene expression
/ allows more copies of mRNA to be made
3 increase amount of gene product / polypeptide in a short period of time
4 to carry out specialised functions in the cell

How gene amplification can occur

Faulty cytokinesis
5 faulty cytokinesis / non-disjunction
6 result in one or more extra sets of chromosomes ending up in a single cell
/ ref. polyploidy
/ ref. duplication of chromosomes sets

Errors in crossing over
7 errors in crossing over / unequal crossing over
8 of non-sister chromatids during prophase I of meiosis
9 even when their homologous gene sequences are **not correctly aligned**

Slippage during DNA replication
10 slippage occur during DNA replication
11 such that the template shifts with respect to the new complementary strand
12 one region of the template strand is copied twice

Significance in development
13 In developing ovum/egg cell, additional copies of the rRNA genes
14 for making enormous number of ribosomes
15 for active protein synthesis (once the egg is fertilized)

9(a) Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]

Starch

Structure
1 amylose is unbranched;
2 polymer of α-glucose molecules linked by α(1, 4) glycosidic bonds

3 hydrophilic hydroxyl groups of glucose residues project into interior of helices;
4 no cross-linking between amylose or glycogen chains
/ no cross-linking to form big bundles

Property and Function
5 **insoluble**;
6 can be stored in large quantities without having any great effect on the water potential of cells
7 and can be prevented from diffusing out of cells

8 individual chains can fold into **compact shape**;
9 can be stored in large quantities (within a cell);

10 **easily hydrolysed** to monosaccharides when required;

Cellulose

Structure
11 unbranched;
12 polymer of β-glucose molecules linked by β(1, 4) glycosidic bonds;

13 alternative units are oriented 180° to each other
/ every other glucose monomer is upside down with respect to the others
14 hydroxyl groups project outwards from each cellulose chain;
15 form H-bonds with neighboring chains
/ cross-linking of chains to form macrofibrils;

Property and Function
16 **great tensile strength**
17 prevents plant cells from bursting when placed in solutions of higher water potential
/ used as cell wall material for structural support in plants;

18 **large intermolecular spaces between macrofibrils**
19 allows passage of water and solute molecules;

9(b) Explain, with two named examples, how the environment may affect the phenotype. [6]

Example 1 [max 3]

1 Himalayan rabbit homozygous for the c^h allele;
2 tyrosinase gene codes for **heat-sensitive enzyme/tyrosinase** that makes melanin;

3 low temp /below 33°C → tyrosinase active / melanin produced;
4 eg. in parts of the body that are cool enough / the extremities / feet, nose, ears and tail → black;

5 high temp /above 33°C → tyrosinase inactive / no melanin produced ;
6 eg. in body regions that are massive enough to conserve a fair amount of heat → white;

7 if a small section of fur is shaved from a Himalayan rabbit, fur will grow back either white or black depending on temp. of environment / Eg. experiment involving ice-pack on shaved rabbit ;
8 just-born rabbits are all white as they had developed at temp. of mother's womb;
9 to produce rabbits with black extremities, the young are kept in the cold;

Example 2 [max 3]

1 In Drosophila homozygous for allele coding for vestigial wing;
2 allele for vestigial wing is recessive to that for long wing;
3 expression of vestigial wing is affected by the temperature at which the insect develops;
4 The allele for vestigial wing is expressed only at low temperatures;
5 develop vestigial wings at 21°C;
6 intermediate wings at 26°C;
7 long wings at 31°C;

Example 3 [max 3]

1 honeybee → drones, queen and workers;
2 drones are males arise from unfertilised eggs;
3 queen and workers are females arise from fertilised eggs;

4 queen and workers have **same amount of genetic material** but phenotypically different;
5 depends on diet of larva;

6 in the first 3 days of hatching, all female larvae feed on royal jelly;
7 after 3 days, those fed with diet consisting of honey/nectar and pollen would develop into workers;
8 workers are sterile, smaller in size and have larger mouthparts and modified legs as compared to the queen;
9 those fed with royal jelly develops into queens as high protein content of royal jelly stimulates formation and maturation of female reproductive system;

Example 4 [max 3]

1 people with functional pancreas/with no type I diabetes have functional genes for insulin uptake;
2 insulin is secreted when blood glucose level increases;
3 overeating of sugary foods for a long period of time causes repeated stimulation of the pancreas;
4 which responds by secreting high levels of insulin;
5 repeated exposure of target cells to large amounts of insulin desensitizes the cells' responsiveness to insulin;
6 result in the target cells failing to take in glucose;
7 resulting in type II diabetes;

9(c) Explain how biogeography and the fossil record support the evolutionary deductions based on homologies. [7]

Biogeography [max 3]

1 the study of the geographic distribution of species;
2 species that are closely related are found closer together;
3 and not found in similar environments far away due to geographical barriers/convergent evolution;
4 e.g. Marsupials are not found anywhere else in the world except in Australia;

5 organisms found close together are evolved from a common ancestor;
6 e.g. Island Biogeography of the Galapagos: Most species of plants and animals living on islands are closely related to neighbouring mainland species;

7 differences suggest evolution from mainland ancestral /descent with modification;
8 different species of finches on Galapagos Islands closely resemble one another;
9 they exhibit homology in terms of shape and size of beak;
10 due to natural selection: Favoured populations with beaks that enabled the birds to acquire their particular food source at each island;

Fossil Records [max 4]

11 Fossils are the preserved remains of organisms.
12 allow scientists to study/compare morphological traits / homologous structures with existing species;
13 coupled with dating of fossils based on radioactive isotopes to determine age of fossil;
14 show that organisms evolved in a chronological manner / historical sequence (fish → amphibians → reptiles → birds and mammals);
15 show new species appearing and others disappearing;
16 reveal ancestral characteristics that have been lost over time;

17 discovery of transitional fossils linking older organisms to modern species supports the idea of descent with modification.

18 provide evidence that **accumulated changes** over long period of time led to **diverse forms** of life present today/diversity of life arose from evolution;

19 DNA/protein could be extracted from fossils and sequences compared;
20 higher molecular homology indicates closer relation;
21 can be used to **prove** that continental drift (land pieces which have drifted apart) leads to divergent evolution;